```
import pandas as pd
df=pd.read_csv('/content/gender_submission.csv')
```

```
df.isnull().sum()
```

	0
PassengerId	0
Survived	0

dtype: int64

```
from sklearn.preprocessing import LabelEncoder

encoder = LabelEncoder()

for col in df.select_dtypes(include='object').columns:
  data[col]=encoder.fit_transform(data[col])
```

```
x=df.iloc[:, 0]
y=df.iloc[:, 1]
```

```
import numpy as np
x=np.array(x)
y=np.array(y)
```

```
x=x.reshape(-1, 1)
y=y.reshape(-1, 1)
```

```
from sklearn.model_selection import train_test_split
```

```
x_train, x_test, y_train, y_test=train_test_split(
    X, Y, test_size=0.2, random_state=42
)
```

```
from sklearn.linear_model import LogisticRegression
```

```
model=LogisticRegression(class_weight='balanced')
```

```
model.fit(x_train,y_train)
```

```
/usr/local/lib/python3.12/dist-packages/sklearn/utils/validation.py:1408: DataConversionWarning: A column-vector y was passe
  y = column_or_1d(y, warn=True)
```

```
▾ LogisticRegression
LogisticRegression(class_weight='balanced')
```

```
y_pred = model.predict(x_test)
```

```
from sklearn.metrics import accuracy_score, confusion_matrix, precision_score, recall_score, f1_score
```

```
accuracy = accuracy_score(y_test, y_pred)
print("Accuracy:",accuracy)
```

```
Accuracy: 0.5238095238095238
```

```
from sklearn.metrics import confusion_matrix

cm = confusion_matrix(y_test, y_pred, labels=[1,0])
print("Confusion Matrix:")
print("TP FP")
print("FN TN")
print(cm)
```

```
Confusion Matrix:
TP FP
FN TN
[[18 16]
 [24 26]]
```

```
precision = precision_score(y_test, y_pred, zero_division=0)
print("Precision:",precision)
```

```
Precision: 0.42857142857142855
```

```
recall = recall_score(y_test, y_pred, zero_division=0)
print("Recall:",recall)
```

```
Recall: 0.5294117647058824
```

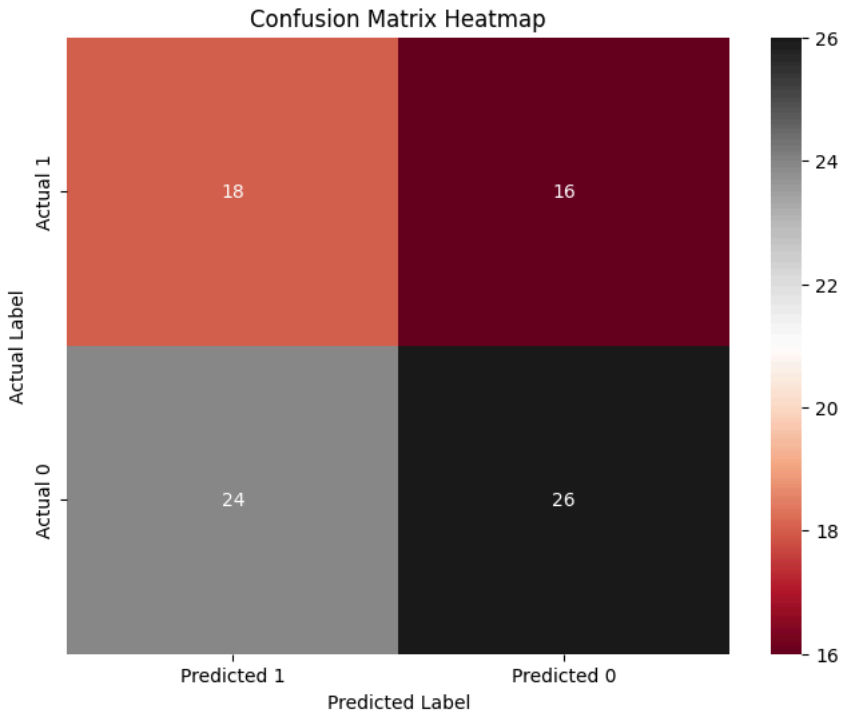
```
f1 = f1_score(y_test, y_pred, zero_division=0)
print("F1 Score:",f1)
```

```
F1 Score: 0.47368421052631576
```

```
import matplotlib.pyplot as plt
import seaborn as sns
```

```
cm = confusion_matrix(y_test, y_pred, labels=[1,0])
```

```
plt.figure(figsize=(8, 6))
sns.heatmap(cm, annot=True, fmt='d', cmap='RdGy',
            xticklabels=['Predicted 1', 'Predicted 0'],
            yticklabels=['Actual 1', 'Actual 0'])
plt.xlabel('Predicted Label')
plt.ylabel('Actual Label')
plt.title('Confusion Matrix Heatmap')
plt.show()
```



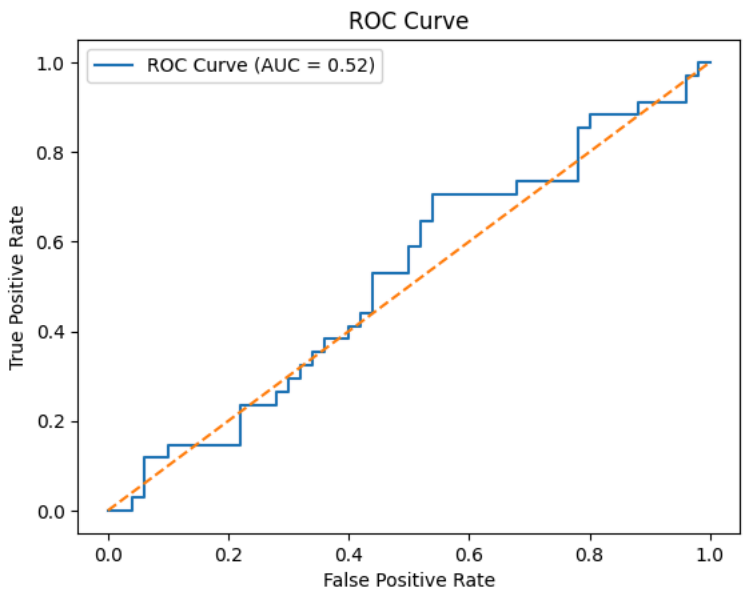

```
from sklearn.metrics import roc_curve, auc
```

```
y_prob = model.predict_proba(x_test)[:, 1]
```

```
fpr, tpr, thresholds = roc_curve(y_test, y_prob)
```

```
roc_auc = auc(fpr, tpr)
print("ROC AUC:",roc_auc)
```

```
ROC AUC: 0.5229411764705882
```

```
plt.figure()
plt.plot(fpr, tpr, label="ROC Curve (AUC = {:.2f})".format(roc_auc))
plt.plot([0,1], [0,1], linestyle='--')
plt.xlabel('False Positive Rate')
plt.ylabel('True Positive Rate')
```

```
plt.title('ROC Curve')
plt.legend()
plt.show()
```